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                                       1        10        20        30        40        50
Corynebacterium_glutamicum_groL     ...ATGGCAAAGCTCATTGCTTTTGACCAGGACGCCCGCGAAGGCATTCTCCGGGGCGTT
Escherichia_coli_groL               ATGGCAGCTAAAGACGTAAAATTCGGTAACGACGCTCGTGTGAAAATGCTGCGCGGCGTA
Thermus_thermophilus_groL           ...ATGGCGAAGATCCTGGTGTTTGACGAGGCTGCGCGCCGCGCCCTGGAGCGCGGGGTC
Bifidobacterium_animalis_groL       ...ATGGCAAAGATCATTGAATACGATGAGGAAGCTCGTCAGGGTATGCTCGCCGGTCTG

                                     60        70        80        90        100       110
Corynebacterium_glutamicum_groL     GACGCTCTGGCAAACGCTGTCAAGGTAACCCTCGGCCCACGCGGCCGTAACGTGGTTCTT
Escherichia_coli_groL               AACGTACTGGCAGATGCAGTGAAAGTTACCCTCGGTCCAAAAGGCCGTAACGTAGTTCTG
Thermus_thermophilus_groL           AACGCGGTGGCCAACGCGGTGAAGGTCACCCTCGGTCCTCGGGGCCGGAACGTGGTCCTG
Bifidobacterium_animalis_groL       GATAAGCTGGCCGATACCGTCAAGGTGACGCTTGGTCCCAAGGGCCGCAACGTCGTGCTT

                                    120       130       140       150       160       170
Corynebacterium_glutamicum_groL     GATAAGGCATTCGGCGGACCTCTGGTCACCAACGACGGTGTCACCATTGCCCGCGACATC
Escherichia_coli_groL               GATAAATCTTTCGGTGCACCGACCATCACCAAAGATGGTGTTTCCGTTGCTCGTGAAATC
Thermus_thermophilus_groL           GAGAAGAAGTTTGGCTCCCCCACCATCACCAAGGACGGGGTGACGGTGGCCAAGGAGGTG
Bifidobacterium_animalis_groL       GACAAGACCTATGGCGCGCCGACCATCACCAACGATGGCGTTTCGATCGCCAAGGAGATC

                                    180       190       200       210       220       230
Corynebacterium_glutamicum_groL     GACCTTGAGGATCCTTTTGAGAACCTCGGTGCGCAGCTGGTGAAGTCCGTTGCTGTTAAG
Escherichia_coli_groL               GAACTGGAAGACAAGTTCGAAAATATGGGTGCGCAGATGGTGAAAGAAGTTGCCTCTAAA
Thermus_thermophilus_groL           GAGCTGGAGGACCACCTGGAGAACATCGGGGCGCAGCTCCTCAAGGAGGTGGCCTCCAAG
Bifidobacterium_animalis_groL       GATCTTGAGGATCCGTTCGAGCGCATTGGTGCCGAACTGGTGAAGGAAGTGGCCAAGCGC

                                    240       250       260       270       280       290
Corynebacterium_glutamicum_groL     ACCAACGACATCGCTGGTGACGGCACCACGACTGCAACTCTGCTTGCTCAGGCACTCATT
Escherichia_coli_groL               GCAAACGACGCTGCAGGCGACGGTACCACCACTGCAACCGTACTGGCTCAGGCTATCATC
Thermus_thermophilus_groL           ACCAACGACGTGGCCGGTGACGGGACCACCACGGCCACCGTCCTCGCCCAGGCCATCGTC
Bifidobacterium_animalis_groL       ACCGATGATGTCGCCGGCGATGGCACCACCACCGCAACTGTACTTGCACAGTCCCTCGTG

                                    300       310       320       330       340       350
Corynebacterium_glutamicum_groL     GCTGAAGGCCTGCGCAACGTTGCTGCTGGCGCAAACCCAATGGAGCTCAACAAGGGTATT
Escherichia_coli_groL               ACTGAAGGTCTGAAAGCTGTTGCTGCGGGCATGAACCCGATGGACCTGAAACGTGGTATC
Thermus_thermophilus_groL           CGGGAGGGCCTGAAGAACGTGGCCGCCGGGGCCAACCCCCTCGCCCTCAAGCGGGGCATT
Bifidobacterium_animalis_groL       CATGAGGGTCTGAAGAACGTGGTGGCCGGATCCAATCCGATTGCCCTGCGCCGCGGCATC

                                    360       370       380       390       400       410
Corynebacterium_glutamicum_groL     TCTGCAGCTGCAGAAAAGACCTTGGAAGAGTTGAAGGCACGCGCAACCGAGGTGTCTGAC
Escherichia_coli_groL               GACAAAGCGGTTACCGCTGCAGTTGAAGAACTGAAAGCGCTGTCCGTACCATGCTCTGAC
Thermus_thermophilus_groL           GAGAAGGCGGTGGAGGCCGCGGTGGAGAAGATCAAGGCCCTCGCCATCCCCGTGGAGGAC
Bifidobacterium_animalis_groL       GAGAAGGCTTCCGACGCTCTCGTCAAGCAGCTTGTCGCTTCCGCGAAGCCGGTGGAGACC

                                    420       430       440       450          460       470
Corynebacterium_glutamicum_groL     ACCAAGGAAATCGCAAACGTCGCTACCGTTTCATCCCG...CGATGAAGTTGTCGGCGAG
Escherichia_coli_groL               TCTAAAGCGATTGCTCAGGTTGGTACCATCTCCGCTAACTCCGACGAAACCGTAGGTAAA
Thermus_thermophilus_groL           CGGAAGGCCATTGAGGAGGTGGCCACCATCTCCGCCAA...CGATCCCGAGGTCGGCAAG
Bifidobacterium_animalis_groL       AAGGAACAGATTGCTGCAACCGCAACGATTTCCGCAGG...TGACCCCGAGGTGGGCGAG

                                       480       490       500       510       520       530
Corynebacterium_glutamicum_groL     ATCGTTGCTGCAGCGATGGAAAAGGTTGGCAAGGACGGTGTCGTCACCGTTGAGGAGTCC
Escherichia_coli_groL               CTGATCGCTGAAGCGATGGACAAAGTCGGTAAAGAAGGCGTTATCACCGTTGAAGACGGT
Thermus_thermophilus_groL           CTGATTGCCGACGCCATGGAGAAGGTGGGGAAGGAGGGCATCATCACCGTTGAGGAGTCC
Bifidobacterium_animalis_groL       AAGATCGCCGAGGCTCTCGACAAGGTCGGACAGGATGGCGTCGTCACCGTCGAAGACAAC

                                       540       550       560       570       580       590
Corynebacterium_glutamicum_groL     CAGTCCATCGAGACTGCTCTCGAGGTCACCGAAGGTATTTCTTTCGACAAGGGCTACCTT
Escherichia_coli_groL               ACCGGTCTGCAGGACGAACTGGACGTGGTTGAAGGTATGCAGTTCGACCGTGGCTACCTG
Thermus_thermophilus_groL           AAGAGCCTCGAGACCGAGCTCAAGTTCGTTGAGGGGTACCAGTTTGACAAGGGGTACATC
Bifidobacterium_animalis_groL       AACCGCTTCGGCCTCGATCTCGATTTCACCGAGGGCATGCGCTTCGACAAGGGTTATATT
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                                      600       610       620       630       640       650
Corynebacterium_glutamicum_groL   TCCCCTTATTTCATCAACGACAACGACACTCAGCAGGCTGTCCTGGACAACCCTGCAGTG
Escherichia_coli_groL             TCTCCTTACTTCATCAACAAGCCGGAAACTGGCGCAGTAGAACTGGAAAGCCCGTTCATC
Thermus_thermophilus_groL         TCCCCCTACTTCGTCACCAACCCCGAGACGATGGAGGCGGTCCTCGAGGACGCCTTCATC
Bifidobacterium_animalis_groL     TCCCCCTACTTCGTCACCAATGCGGAAGACCAGACGGCCGTGCTCGACGACCCGTACATT

                                      660       670       680       690       700       710
Corynebacterium_glutamicum_groL   CTGCTTGTTCGCAACAAGATTTCTTCCCTCCCAGACTTCCTCCCATTGCTGGAGAAGGTT
Escherichia_coli_groL             CTGCTGGCTGACAAGAAAATCTCCAACATCCGCGAAATGCTGCCGGTTCTGGAAGCTGTT
Thermus_thermophilus_groL         CTCATCGTGGAGAAGAAGGTCTCCAACGTCCGTGAGCTCCTCCCCATCCTGGAGCAGGTG
Bifidobacterium_animalis_groL     CTGCTCACCTCGAGCAAGGTCTCCTCGCAGCAGGATGTGGTCCACATCGCCGAACTGGTG

                                      720       730       740       750       760       770
Corynebacterium_glutamicum_groL   GTGGAGTCCAACCGTCCTTTGCTGATCATCGCAGAAGACGTCGAGGGCGAGCCTTTGCAG
Escherichia_coli_groL             GCCAAAGCAGGCAAACCGCTGCTGATCATCGCTGAAGATGTAGAAGGCGAAGCGCTGGCA
Thermus_thermophilus_groL         GCCCAGACGGGCAAGCCCCTCCTGATCATCGCCGAGGACGTGGAGGGCGAGGCTTTGGCC
Bifidobacterium_animalis_groL     ATGAAGACCGGCAAGCCGTTGCTGATCGTGGCCGAGGATGTCGATGGCGAGGCACTGCCC

                                      780       790       800       810       820       830
Corynebacterium_glutamicum_groL   ACCCTGGTTGTGAACTCCATCCGCAAGACCATCAAGGTCGTTGCAGTGAAGTCCCCTTAC
Escherichia_coli_groL             ACTCTGGTTGTTAACACCATGCGTGGCATCGTGAAAGTCGCTGCGGTTAAAGCACCGGGC
Thermus_thermophilus_groL         ACCCTGGTGGTGAACAAGCTCCGGGGCACCCTGAGCGTGGCCGCGGTGAAGGCTCCCGGC
Bifidobacterium_animalis_groL     ACGCTGATTCTCAACAACATCCGTGGCACCTTCAAGTCCTGCGCCGTCAAGGCTCCGGGC

                                      840       850       860       870       880       890
Corynebacterium_glutamicum_groL   TTCGGTGACCGACGCAAGGCGTTCATGGATGACCTGGCTATTGTCACCAAGGCAACTGTC
Escherichia_coli_groL             TTCGGCGATCGTCGTAAAGCTATGCTGCAGGATATCGCAACCCTGACTGGCGGTACCGTG
Thermus_thermophilus_groL         TTCGGTGACCGCAGGAAGGAGATGCTCAAGGACATCGCGGCCGTCACCGGCGGCACCGTG
Bifidobacterium_animalis_groL     TTCGGTGACCGCCGCAAGGCCATGCTGCAGGATATGGCGATTCTGACCGGCGGTCAGGTC

                                      900       910       920       930       940       950
Corynebacterium_glutamicum_groL   GTGGATCCAGAAGTGGGCATCAACCTCAACGAAGCTGGCGAAGAAGTTTTCGGTACCGCA
Escherichia_coli_groL             ATCTCTGAAGAGATCGGTATGGAGCTGGAAAAAGCAACCCTGGAAGACCTGGGTCAGGCT
Thermus_thermophilus_groL         ATCTCCGAGGAGCTCGGCTTCAAGCTGGAGAACGCCACCCTCTCCATGCTGGGCCGGGCC
Bifidobacterium_animalis_groL     GTCTCGGAGGATCTGGGCCTCAAGCTCGATTCCATCGACCTGTCCGTGTTCGGCACCGCC

                                      960       970       980       990      1000      1010
Corynebacterium_glutamicum_groL   CGCCGCATCACCGTTTCCAAGGACGAAACCATCATCGTTGATGGTGCAGGTTCCGCAGAA
Escherichia_coli_groL             AAACGTGTTGTGATCAACAAAGACACCACCACTATCATCGATGGCGTGGGTGAAGAAGCT
Thermus_thermophilus_groL         GAGCGGGTGCGCATCACCAAGGACGAGACCACCATCGTGGGCGGCAAGGGCAAGAAGGAG
Bifidobacterium_animalis_groL     AAGAAGGTCATTGTCTCCAAGGATGAGACCACCATCGTGTCCGGTGGCGGCTCCAAGGAG

                                     1020      1030      1040      1050      1060      1070
Corynebacterium_glutamicum_groL   GACGTTGAAGCACGTCGCGGCCAGATCCGTCGCGAAATCGCCAACACCGATTCCACCTGG
Escherichia_coli_groL             GCAATCCAGGGCCGTGTTGCTCAGATCCGTCAGCAGATTGAAGAAGCAACTTCTGACTAC
Thermus_thermophilus_groL         GACATTGAGGCCCGGATCAACGGCATCAAGAAGGAGCTGGAGACCACCGACAGCGAGTAC
Bifidobacterium_animalis_groL     GACGTGGCCGCACGCGTCGCCCAGATTCGCGCCGAGATCGAGAAGACCGATTCCGATTAC

                                     1080      1090      1100      1110      1120      1130
Corynebacterium_glutamicum_groL   GATCGCGAAAAGGCAGAAGAGCGTTTGGCTAAGCTCTCCGGTGGTATTGCTGTCATCCGC
Escherichia_coli_groL             GACCGTGAAAAACTGCAGGAACGCGTAGCGAAACTGGCAGGCGGCGTTGCAGTTATCAAA
Thermus_thermophilus_groL         GCCCGGGAGAAGCTCCAGGAGCGCCTGGCGAAGCTCGCGGGTGGCGTGGCCGTGATCCGG
Bifidobacterium_animalis_groL     GATCGTGAGAAGCTGCAGGAGCGTTTGGCTAAGCTGGCAGGTGGCGTCGCCGTCATCAAG

                                     1140      1150      1160      1170      1180      1190
Corynebacterium_glutamicum_groL   GTTGGTGCAGCAACTGAAACCGAAGTCAACGACCGCAAGCTGCGTGTCGAAGATGCCATC
Escherichia_coli_groL             GTGGGTGCTGCTACCGAAGTTGAAATGAAAGAGAAAAAAGCACGCGTTGAAGATGCCCTG
Thermus_thermophilus_groL         GTGGGCGCCGCCACGGAGACCGAGCTCAAGGAGAAGAAGCACCGCTTTGAGGACGCCCTG
Bifidobacterium_animalis_groL     GTCGGCGCAGCCACCGAGGTCGAGGCCAAGGAGCGCAAGCACCGCATTGAAGATGCCGTG
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                                       1200      1210      1220      1230      1240      1250
Corynebacterium_glutamicum_groL     AACGCTGCTCGCGCAGCAGCACAAGAAGGCGTTATCGCTGGTGGCGGTTCCGCTTTGGTT
Escherichia_coli_groL               CACGCGACCCGTGCTGCGGTAGAAGAAGGCGTGGTTGCTGGTGGTGGTGTTGCGCTGATC
Thermus_thermophilus_groL           AACGCCACCCGGGCCGCGGTGGAGGAGGCATCGTGCCCGGCGGCGGCGTGACCCTCCTC
Bifidobacterium_animalis_groL       CGCAACGCCAAGGCCGCCATCGAAGAGGGTCTGGTTCCGGGCGGCGGCGTCGCACTGGTG

                                       1260      1270      1280      1290      1300      1310
Corynebacterium_glutamicum_groL     CAGATCGCTGAGACTCTGAAGGCTTACGCCGAAGAGTTCGAAGGCGACCAGAAGGTCGGC
Escherichia_coli_groL               CGCGTAGCGTCTAAACTGGCTGACCTGCGTG...GTCAGAACGAAGACCAGAACGTGGGT
Thermus_thermophilus_groL           CGGCCATCAGCGCCGTGGAGGAGCTCATCAAGAAGCTGGAGGGCGACGAGGCCACGGGC
Bifidobacterium_animalis_groL       CAGGCTGCCGAAAAGGTTGAGAAGGACTT...CAACCTTGAAGGCGACGAGGCCACCGGT

                                       1320      1330      1340      1350      1360      1370
Corynebacterium_glutamicum_groL     GTTCGCGCACTGGCTACTGCTTTGGGCAAGCCAGCGTACTGGATCGCCTCCAACGCAGGT
Escherichia_coli_groL               ATCAAAGTTGCACTGCGTGCAATGGAAGCTCCGCTGCGTCAGATCGTATTGAACTGCGGC
Thermus_thermophilus_groL           GCCAAGATCGTGCGCCGGGCCCTGGAGGAACCCGCCCGCCAGATCGCCGAGAACGCCGGG
Bifidobacterium_animalis_groL       GCCGCAATCGTCTTCTCGGGCATCGAGGCTCCGATTAAGCAGATCGCCGAAAATGCAGGT

                                       1380      1390      1400           1410      1420
Corynebacterium_glutamicum_groL     CTTGACGGCTCTGTTGTTGTTGCACGCACTGCT......GCTCTGCCAAACGGCGAGGC
Escherichia_coli_groL               GAAGAACCGTCTGTTGTTGCTAACACCGTTAAA......GGCGGCGACGGCAACTACGGT
Thermus_thermophilus_groL           TACGAGGGCTCCGTCATCGTCCAGCAGATCCTGGCCGAGACCAAGAACCCCCGCTACGGC
Bifidobacterium_animalis_groL       CTCTCTGGCGCCGTGGTGATCGACAAGGTTCGTTCCCTG......CCTGAGGGTGAGGGC

                                   1430      1440      1450      1460      1470      1480
Corynebacterium_glutamicum_groL     TTCAACGCTGCAACTTTGGAATACGGAAACCTGATCAACGACGGTGTCATCGACCCAGTC
Escherichia_coli_groL               TACAACGCAGCAACCGAAGAATACGGCAACATGATCGACATGGGTATCCTGGATCCAACC
Thermus_thermophilus_groL           TTCAACGCCGCCACCGGGGAGTTCGTGGACATGGTGGAGGCGGGCATCGTTGACCCCGCC
Bifidobacterium_animalis_groL       TTCAATGCGGCAACCGACACCTATGAGGATCTCATGGCCGCCGGTGTGACCGATCCTGTC

                                   1490      1500      1510      1520      1530      1540
Corynebacterium_glutamicum_groL     AAGGTCACCCATTCCGCAGTAGTGAATGCAACCTCTGTTGCACGCATGGTTCTGACCACT
Escherichia_coli_groL               AAAGTAACTCGTTCTGCTCTGCAGTACGCAGCTTCTGTGGCTGGCCTGATGATCACCACC
Thermus_thermophilus_groL           AAGGTGACCCGCTCTGCCCTGCAGAACGCCGCCTCCATCGGCGCCCTCATCCTCACCACC
Bifidobacterium_animalis_groL       AAGGTGACTCGTTCTGCTTTGCAGAATGCGGCCTCCATCGCGGGTCTGTTCCTGACGACC

                                   1550      1560      1570      1580      1590      1600
Corynebacterium_glutamicum_groL     GAGGCTTCTGTTGTTGAGAAGCCTGCAGAAGAAGCAGCCGATGCACATGCAGGACATCAT
Escherichia_coli_groL               GAATGCATGGTTACCGACCTGCCGAAAAACGATGCAGCTGACTTAGGCGCTGCTGGCGGT
Thermus_thermophilus_groL           GAGGCGGTGGTGGCGGAGAAGCCCGAGAAGAAGGAGTCCACCCCCGCCTCCGCGGGCGCC
Bifidobacterium_animalis_groL       GAAGCGGTTGTCGCCAACAAGCCGGAGCCGGCAGCGGCACCGGCCGCCGGCCAGG.....

                                   1610
Corynebacterium_glutamicum_groL     CACCACTAA...........................
Escherichia_coli_groL               ATGGGCGGCATGGGTGGCATGGGCGGCATGATGTAA
Thermus_thermophilus_groL           GG...GGACATGGACTTCTAA...............
Bifidobacterium_animalis_groL       .......ACATGGGTTACTGA...............
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